



The auto-search feature has been disabled based on user feedback. Enter a search term/phrase and click "Search" to begin.

[Back to Results](#)

Leading-edge slat optimization for maximum airfoil lift

A numerical procedure for determining the position (horizontal location, vertical location, and deflection) of a leading edge slat that maximizes the lift of multielement airfoils is presented. The structure of the flow field is calculated by iteratively coupling potential flow and boundary layer analysis. This aerodynamic calculation is combined with a constrained function minimization analysis to determine the position of a leading edge slat so that the suction peak on the nose of the main airfoil is minized. The slat position is constrained by the numerical procedure to ensure an attached boundary layer on the upper surface of the slat and to ensure negligible interaction between the slat wake and the boundary layer on the upper surface of the main airfoil. The highest angle attack at which this optimized slat position can maintain attached flow on the main airfoil defines the optimum slat position for maximum lift. The design method is demonstrated for an airfoil equipped with a leading-edge slat and a trailing edge, single-slotted flap. The theoretical results are compared with experimental data, obtained in the Ames 40 by 80 Foot Wind Tunnel, to verify experimentally the predicted slat position for maximum lift. The experimentally optimized slat position is in good agreement with the theoretical prediction, indicating that the theoretical procedure is a feasible design method.

Document ID 19790018929

Acquisition Source
Legacy CDMS

Document Type
Technical Memorandum (TM)

Authors
Olson, L. E.
(NASA Ames Research Center Moffett Field, CA, United States)

Mcgowan, P. R.

(Computer Sci. Corp. Mountain View, Calif., United States)

Guest, C. J.

(Computer Sci. Corp. Mountain View, Calif., United States)

Date Acquired September 3, 2013

Publication Date
July 1, 1979

Subject Category
Aerodynamics

Report/Patent Number
A-7753 **NASA-TM-78566**

Report Number: A-7753

Report Number: NASA-TM-78566

Accession Number
79N27100

Funding Number(s)
PROJECT: RTOP 505-06-31

Available Downloads

Name	Type	
19790018929.pdf	STI	  

Related Records

There are no records associated with this record.



No Preview Available

Distribution	Public
Limits	
Copyright	Work of the US Gov. Public Use Permitted.



[No Fear Act](#) | [Freedom of Information Act](#) | [Office of the Inspector General](#) | [Agency Financial Reports](#)
[NASA Web Privacy Policy and Important Notices](#) | [NASA Guidelines for Quality of Information](#)
[Disclaimers, Copyright, Terms of Use](#) | [Accessibility](#) | [NASA OCIO](#) | [USA.gov](#) | [NASA.gov](#)
[Contact Us](#) | [Site Map](#)



NASA Official: JoAnne R. Calhoun
Site Curator: STI Compliance and Distribution Services
Last Modified: June 9, 2026